



The Technology Value Stream

Defining Lead Time vs Processing Time
Taking Deployment Lead Times from Months to Minutes

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Mod.1.2

Building the Strategic Capability

“Improving daily work is more important than doing daily work.” – Gene Kim



LEAD TIME VS. PROCESSING TIME



Lead Time

- Starts when the **customer request** is made and ends when the request is fulfilled.
- It represents the total elapsed time as perceived by the customer.



Processing Time

- Known as "touch time." The clock starts only when **work begins** on the request.
- It doesn't count the time the work spends waiting in queues.



Hidden Cost of Queues

90% Lead Time Waste

Waiting vs. Working

- In most traditional organizations, the actual work (processing time) is a tiny fraction of the total lead time.
- Delays do not come from the work itself, but from the idle time where tickets sit in queues waiting for approvals, environments, or manual testing.



The Struggle:

(Months of Delay)

Monolithic Systems:

Tightly coupled architectures prevent independent deployment.

Manual Testing:

Heavy reliance on manual QA cycles that take weeks.

Approval Gates:

Multiple bureaucratic layers and Change Review Boards.

Scarce Environments:

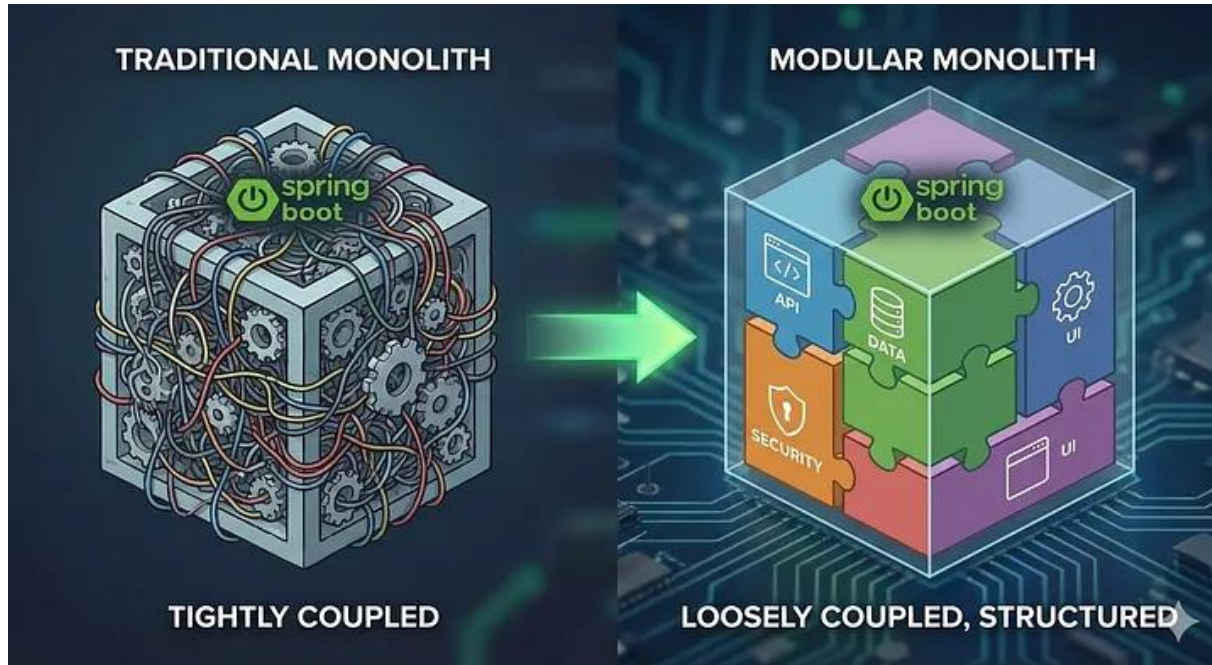
Long wait times for integration and test environments.



Our DevOps Ideal

Deployment Lead Times Measured in Minutes

Architecture for Speed



- **Traditional Monolith (Tightly Coupled)**
 - Failures are global and contained; one small error can break the entire system.
- **Modular Monolith (Loosely Coupled, Structured)**
 - Small teams work with high autonomy
 - Continually checking small code changes into version control repositories daily; Automated flows and fast feedback result in high confidence
 - Failures are isolated and easily corrected



Performance Benchmark

Shorter lead times correlate with higher organizational agility

Elite Performers

Lead Time: Under 1 Hour

High Performers

Lead Time: 1 Day – 1 Week

Medium Performers

Lead Time: 1 Week – 1 Month

Low Performers

Lead Time: 1 – 6+ Months

Resources (DevOps Handbook, Online Content, Slides/Templates)

Source Title and Link	Slide(s) Referenced
The DevOps Handbook (pg. 49-58): How to Create World-Class Agility, Reliability, & Security in Technology Organizations	Slides 3, 4, 5, 7
Medium: 5-Minute DevOps: The Strategic Reasons for DevSecOps	Slide 2
Octopus Deploy: Why are there so many definitions of Lead Time?	Slide 3
Azure DevOps: Configure lead time and cycle time widgets	Slides 3, 4
Humanitec: Lead Time – a Key Metrics in DevOps	Slide 4
Medium/Stackademic: Spring Boot is Becoming a Modular Monolith	Slide 7
Axify: Deployment Time: What it is. How to Measure & Improve	Slide 8
DX: What are DORA Metrics? Complete guide to measuring DevOps performance	Slide 8
Power Point Templates: Bright business presentation, Classic corporate teach a course slides	Slides 1 - 10



Thank You

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CSD 380 – Module 1.2



Professor Mazhata

